

NAINA OS -- VOLUME 9

DEVELOPER PLATFORM & NAINA SDK (DPN SDK v1.0)

"An operating system becomes successful when other people can build on it."

DOCUMENT METADATA

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Core Focus:	Subsystem SDKs * Marketplace * Portal * 15-Volume Roadmap

This document contains formal engineering specifications for the NAINA OS SDK Suite, BaseAgent Interface, Skill Manifests, AI Plugin Generator Engine, Marketplace, and 15-Volume Roadmap.

NAINA OS -- Developer Platform & NAINA SDK (DPN SDK v1.0)

Volume 9: Developer Platform, Subsystem SDKs, AI Plugin Generator & 15-Volume Roadmap

“An operating system becomes successful when other people can build on it.”

Document Revision History

Date	Revision	Author	Description of Changes
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2026-07-31	v1.0.0	Chief Systems Architect	Release of Volume 9 -- Developer PI
2026-07-30	v0.9.0	Developer SDK Team	Technical specifications for Agent,

FLAGSHIP SUBSYSTEM: AI Plugin Generator Engine (APGE)

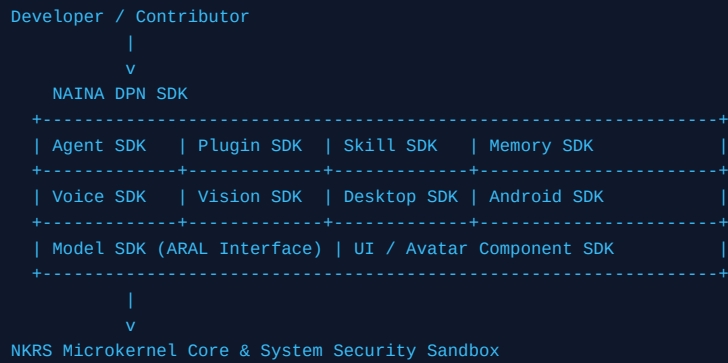
1. Conversational Plugin Generation Architecture

The AI Plugin Generator Engine (APGE) enables developers or end-users to generate complete, production-ready NAINA OS plugins using natural language:

```
User Prompt: "Naina, create a plugin that connects to my DJI drone."
|
v
[NAINA] NAINA Companion Engine
|
v
[AI] APGE Goal Planner & Code Synthesis Engine
|
+-- Generates plugin/manifest.yaml (Capability & Tool Claims)
+-- Generates plugin/plugin.py (Python SDK Async Implementation)
+-- Generates plugin/tests/test_drone.py (Pytest Unit Tests)
+-- Generates plugin/docs/README.md & Architecture Diagram
+-- Applies Security Capability Token Checks
|
v
Developer Interactive Code Review & Approval Gate
|
v
Plugin Installed to NAINA OS Local Extension Sandbox
```

DOCUMENT 09.1 -- NAINA SDK Suite Overview

The NAINA OS SDK (DPN SDK) is the official developer abstraction layer extending system functionality:



DOCUMENT 09.2 -- Agent SDK Specification

All domain agents implement the universal BaseAgent abstract class:

```

# NAINA OS Base Agent Abstract Specification (Python SDK)
from abc import ABC, abstractmethod
from typing import Dict, Any

class BaseAgent(ABC):
    def __init__(self, agent_id: str, capability_token: str):
        self.agent_id = agent_id
        self.capability_token = capability_token
        self.is_running = False

    @abstractmethod
    async def initialize(self) -> bool:
        """Loads configuration and establishes IPC event subscriptions."""
        pass

    @abstractmethod
    async def execute(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        """Executes the assigned domain task contract."""
        pass

    @abstractmethod
    async def pause(self) -> bool:
        """Temporarily yields resource tokens and pauses queue processing."""
        pass

    @abstractmethod
    async def resume(self) -> bool:
        """Resumes active queue processing."""
        pass

    @abstractmethod
    async def shutdown(self) -> bool:
        """Flushes buffers and releases system locks."""
        pass

    @abstractmethod
    async def health(self) -> Dict[str, Any]:
        """Returns heartbeat telemetry and error counters."""
        pass

```

DOCUMENT 09.3 -- Skill SDK Specification

Skills package multi-agent workflows into reusable operational blueprints:

```
# Standard Skill Manifest Schema (skill.manifest.yaml)
id: "skill.morning_routine"
name: "Morning Briefing Routine"
version: "1.0.0"
description: "Ingests weather, calendar, GitHub PRs, and synthesizes a morning voice briefing."
author: "Jonathan <jonathan@nainaos.org>"
permissions:
  - "CAP_NET_HTTP"
  - "CAP_OBSIDIAN_READ"
  - "CAP_VOICE_TTS"
required_agents:
  - "agent.browser"
  - "agent.obsidian"
  - "agent.voice"
required_tools:
  - "weather_get_current"
  - "google_calendar_list_events"
  - "github_list_prs"
timeout_seconds: 60
```

DOCUMENT 09.4 -- Plugin SDK Architecture

Every NAINA OS plugin follows a standardized directory structure:

```
my-dji-drone-plugin/
+-- manifest.yaml      # Plugin metadata & required capability tokens
+-- plugin.py          # Primary Python plugin implementation
+-- README.md          # Auto-generated documentation & usage guide
+-- assets/            # Icon badges, diagrams & visual assets
+-- tests/             # Unit & integration test suites
+-- docs/              # API reference specs
```

DOCUMENT 09.5 -- Memory SDK Specification

Third-party plugins access memory strictly through permission-controlled proxy APIs:

```
// Permission-Gated Memory SDK (TypeScript Contract)
export class MemorySDK {
  constructor(private capabilityToken: string) {}

  async search(query: string, limit: number = 5): Promise<Array<Record<string, unknown>>> {
    // Verifies CAP_OBSIDIAN_READ capability before querying pgvector
    return [];
  }

  async store(content: string, metadata: Record<string, unknown>): Promise<string> {
    // Verifies CAP_OBSIDIAN_WRITE capability
    return "memory_id_1092a";
  }

  async addRelationship(sourceEntity: string, targetEntity: string, relation: string): Promise<bool> {
    // Inserts triple edge into Knowledge Graph
    return true;
  }
}
```

DOCUMENT 09.6 through 09.8 -- Voice, Vision & Model SDKs

- * Voice SDK: Exposes voice.say(), voice.listen(), voice.stream(), voice.interrupt() without exposing raw microphone devices.
- * Vision SDK: Unified interface across OpenClaw and Qwen-VL: vision.capture(), vision.locate_ui(), vision.ocr().
- * Model SDK (ARAL): Unified abstraction across all LLMs: model.chat(), model.embed(), model.vision(), model.stream(), model.tokens().

DOCUMENT 09.9 & 09.10 -- Extension Marketplace & Auto-Documentation

```
flowchart TD
    Dev[Developer Submits Plugin] --> Pipeline[Automated CI/CD Verification Pipeline]
    Pipeline --> S1[Stage 1: Manifest & Capability Token Verification]
    Pipeline --> S2[Stage 2: Static Security & Malware Audit]
    Pipeline --> S3[Stage 3: Auto-Generate Docs & OpenAPI Specs]
    Pipeline --> S4[Stage 4: Pytest Sandbox Integration Suite]
    S4 -- All Passed --> Marketplace[Publish to NAINA OS Extension Marketplace]
```

NAINA DEVELOPER PORTAL (`docs.nainaos.org`)

The NAINA Developer Portal serves as the central hub for the developer ecosystem:

- * API References: Auto-generated interactive OpenAPI / TypeDoc documentation.
- * SDK Guides: Tutorials for Python, TypeScript, and Rust agent development.
- * Marketplace Directory: Discover, audit, and install verified community plugins.

MASTER ARCHITECTURAL ROADMAP (15-VOLUME SERIES)

NAINA OS COMPLETE 15-VOLUME MASTER ROADMAP			
Volume	Architectural Domain	Status	
Volume 1	Foundation & Architecture Blueprint	APPROVED	
Volume 2	Microkernel & Runtime (NKRS v1.0)	APPROVED	
Volume 3	AI Integration & ARAL Framework (AIIF v1.0)	APPROVED	
Volume 4	Memory & Knowledge Engine (MKIE v1.0)	APPROVED	
Volume 5	Desktop & Device Interaction (DMCIF v1.0)	APPROVED	
Volume 6	Voice OS & Zero-Trust Security (VOSP v1.0)	APPROVED	
Volume 7	Cognitive Architecture & Reasoning (CARF)	APPROVED	
Volume 8	Multi-Agent Civilization (MACF v1.0)	APPROVED	
Volume 9	Developer Platform & SDK (DPN SDK v1.0)	APPROVED	
Volume 10	Vision, Spatial Computing & Computer Use	PLANNED (v1.1)	
Volume 11	Security, Governance & Audit Standard	PLANNED (v1.1)	
Volume 12	Enterprise DevOps, Infrastructure & Cloud	PLANNED (v1.2)	
Volume 13	UI/UX Design System & Avatar Engine	PLANNED (v1.2)	
Volume 14	Robotics, ROS2 & Edge Physical Computing	PLANNED (v2.0)	
Volume 15	Future Quantum & Post-LLM Research	PLANNED (v2.0)	

End of NAINA OS Developer Platform & NAINA SDK -- Volume 9 (Documents 09.1 - 09.10, APGE & 15-Volume Roadmap)